

Hudson Carbon Research Narrative

Stone House Farm History: Geologic, Economic & Social

Stone House Farm is located in the Hudson River Basin in land that once was the bottom of the Devonian Sea. At the time the heavy clay particles settled to the lower points at the lake bottom and the lighter silt and sand particles settled in the shallower edges of the lake. This million-year history is still relevant today as the soils at Stone House Farm vary greatly from heavy clay on much of the lower land on the farm and silty loam on the higher fields. About 65 percent of the Stone House Farm farmland is comprised of heavy clay soil and the remaining 35 percent is comprised of better drained, silt and sand-loam soils. These baseline characteristics make much of the farm difficult to manage in wet weather and difficult to maintain consistent harvests, especially with annual crops. About 520 acres of prime cropland on the farm that make consistent grain yields and have the potential to grow a wide variety of specialty crops in addition to grain.

Stone House Farm is comprised of several former farms, some dating back to the early 1700's. Over the past three centuries the land has supported many crops. The first European settlers started with diversified subsistence farming and then settling into mostly livestock (dominated by sheep in the 1800's), grain, and some orchards. With the advent of refrigeration in the 1900's the Hudson Valley became a major dairy and apple producing region. By the 1940's small dairies and orchards came to dominate the area with some vegetable farms as well. There was a thriving regional agrarian economy through the 1970's. In the early 1980's food commodity prices collapse and a long painful farm crisis ensued. It was during this time and in response to this crisis that Peggy McGrath Rockefeller founded the American Farmland Trust and began purchasing distressed dairies here in Livingston and Germantown, NY with the goal of establishing a viable model for agriculture in the region that hinged on land conservation.

The result was establishment of the beautiful Stone House Farm. From the onset, the Stone House Farm infrastructure was comprised of various dairy infrastructure built between 1900 and 1975. This aging infrastructure represented a previous era and was not applicable to the grain farming and beef cattle breeding that Stone House Farm began in earnest in 1981. The Home Farm, previously owned by the Vorhees family, did have a large shop, three grain bins and a small feed mill already built when the farm was purchased in 1981, from that time on this farm has served as the home base for the Stone House Farm, housing the farm offices, grain facilities, repair shop, and core cattle handling infrastructure. The Clum Farm is a secondary base of operations.

Additional Farm Properties within Stone House Farm Scope of Management

In addition to the land comprising Stone House Farm, two other adjacent properties were under the management of the farm during the transition, and throughout the management tenure of Ben Dobson. These properties were the Jahcoln Farm, and Old Mud Creek Farm. The Jahcoln Farm, a former dairy farm consisting of approximately 150 acres of mostly heavy silt & clay loam soils, was acquired by Dirt Capital Partners *in 2013*, and leased to Stone House as cropland in 2014. Because of soil texture and lack of drainage, it proved difficult to grow successful annual cash crops on the majority of this farm, with the exception of 20 acres of lighter, well-drained soils that produced modestly successful cash crops. The farm was eventually seeded down entirely to perennial pasture/hay by the fall of 2022.

Old Mud Creek Farm is located immediately north of the Jahcoln Farm, separated only by rural road, and connected via the Old Mud Creek, that originates from a spring within the Jahcoln property, and passes under the road via a culvert. It consists of approximately 200 acres of open farmland, with soils ranging from heavy silt & clay, to lighter sandy loams. Old Mud Creek Farm is unique in that it's most interesting phase of management was during the 1990's, when it served as the Northeast Research Station for Syngenta. The function of the research station was to test products developed by the agrochemical company for regulatory approval within the region. As such, the farm had a diverse mix of cropping systems, ranging from annual commodity crop production, to specialty vegetable & fruit production, including orchards. Prior to this phase, the farm was typical of most family farms in the region, producing dairy, grain & forage crops. The land changed hands periodically, through sales & leases, and was finally purchased and conserved by Abby in 2015, with the assistance of Scenic Hudson, a region land conservation & environmental advocacy organization. The farm was then wrapped into the larger organic transition of Stone House Farm, and three of the original 12 research sites were located across Old Mud Creek.

The Stone House Farm management team also undertook to manage an organic transition across three other farms in the area: The Esselstyn, Runyon, and Thorne Farms. This pushed the total acreage under management to ~3000 during the organic transition phase. Two of the three farms, Esselstyn & Runyon, were relatively distant from the main farm operation, and the timing and logistical challenges made managing these farms difficult. Management was ultimately transferred to another organic farm operation once the 3-year organic transition period was complete. Most of the Thorne's land remained under SHF management, until they endeavored to set up a farm of their own, which operates today as Gulden Farm, a 100% grass-fed beef operation.

Historical Practices

Cattle and Conventional Farming

When Stone House Farm was first established, it focused on breeding and advancing the Simmental cattle breed, and advancing grain production. Many hedgerows were cleared, and many of the fields across the farm were combined into larger open parcels to maximize efficiency and production. Tile drains were installed across the farm and drainage ditches dug to move the water from tile drains off the farm to drainage swales. Fences were constructed across the farm, cattle handling and breeding facilities were constructed at the Home Farm,

and the grain facilities were expanded to include a grain handling leg with a loading and unloading area.

Through the 1980's and up until the late 1990's the farm used conventional tillage and synthetic inputs to maximize grain yields. The farm was consistently the highest yielding grain operation in the county and was responsible for introducing soybean production to the area. Cattle were originally managed in a set-stock system, mainly to more efficiently manage breeding bulls and cows. During this time Stone House Farm developed a reputation as a top tier grain producer with terrific equipment operators. The grain production system and cattle enterprise were all developed under the management of Jeff McCallister and with Don Homer assisting Peggy.

In 1990 John Walston joined the operation and furthered the farms cattle breeding program and grain production. John brought the Stone House reputation for Simmental cattle to the national stage and was often a state finalist in corn yield competitions, while also producing high yields in soybeans. In 1991 John was joined by Rick Burd. Rick is a tireless worker who was instrumental in the efficient management of Stone House Farm and the high crop yields.

In the mid 1990's Peggy acquired the Marty Schroeder Farm and the Sirmone Farm, bringing the acreage of the operation to its current total.

No-Till Rotation

In the mid 1990's, John and Rick began the conversion at Stone House Farm to a no-till operation. This was largely in response to the heavy clay soils on the farm and the cost savings represented by no till. The transition was a success, with crop yields remaining stable and soil organic matter in the top layer of the soil increasing by as much as 1%.

Through the 1990's the operation was also leasing large tracts of land in Claverack, and Livingston grain production acreage reached a high of 1,800 acres at this time. The use of conventional inputs remained high to support high crops yields.

The operation also went through a transition in 1990's with it's cattle program, with most breeding activities moving to the Hudson Pines operation in Tarrytown, NY, and Stone House Farm transitioning most of its cattle program to pasturing steers for the grazing season. By the second half of the 1990's Stone House had settled into a solid rhythm and became very efficient under John and Rick's oversight.

Organic Transition

In 2011 Abby acquired the land that is now home to the Churchtown Dairy. This farm, located in Churchtown, NY, was already seeded to hay and pasture and organic transition began simply with the elimination of synthetic inputs. In 2012 construction began at the Churchtown Dairy.

At this time PMR was actively exploring options for an organic transition, and meeting with the Stone House Farm management team and other institutions and individuals to develop a transition plan. I joined PMR as an advisor to the organic transition in early 2013 and participated in a series of listening sessions with regional farmers, conservation organizations, non-profit organizations, and food businesses. These listening sessions made it clear that there was a strong local demand for organic and non-GMO grains and that Stone House Farm could genuinely provide a key input to the burgeoning local farm and food movement. The transition to organic grain also made the most sense due to the farms' infrastructure, equipment and team.

Throughout 2013, I spent two to three days a week meeting with John and Rick to best understand practices used, their ideas for organic transition, and to discuss various approaches and options for the transition to organic production.

During this time, I explored all of the farm, observing crop growth, soil types, and building a transition plan to follow the harvest of the final conventional crops. I feel that Rick and John did a good job of setting aside their conventional practices and implementing an organic program in a measured and strategic manner. Rick was particularly excited about no-till organic practices and did a good job implementing them.

Regenerative Organic Practices & Transition History

Beginning in 2013, organic practices and crop rotations were deployed and developed at Stone House Farm (including Esselstyns, Runyons, & Thornes), Old Mud Creek Farm, and The Churchtown Dairy. In addition to grains & forage production, an intensively-managed, multi-paddock grazing rotation was implemented at Stone House Farm for grass-fed beef production, and a multi-paddock dairy grazing rotation began at Churchtown. The following section highlights the practices used and rotations developed between 2013 and 2020, across the roughly 3000 acres under management.

2013

In 2013 Stone House Farm planted and harvested conventional corn, soy, and wheat crops along with some hay. The last herbicides and synthetic fertilizers were applied in June 2013.

In anticipation of the organic transition, oats were planted with an under-sowing of clover on the Hermitage field HM1 and rye vetch cover crops were planted at Wagner Tinklepaugh fields WA1 and WA5, rye and Austrian winter peas were seeded at Home Farm fields HF 2A and HF 13.

Winter wheat and rye (for grain and cover crop) were seeded following all soybean fields in order to build organic matter, generate revenue, and establish a window of time after the following season (2014) harvests to establish cover crops. Rye was planted following the corn stubble in late October.

The plan was to establish a three to four year crop rotation designed to maximize organic no-till practices and organic grain production.

2014

The spring of 2014 marked the first season of organic crops. In advance of the season the following equipment was purchased:

1. Case 6140 combine with corn harvest head and small grain head
2. Case 185 Tractor
3. 31' roller crimper

4. Turbo Till vertical tillage implement
5. 8 bottom moldboard plow
6. 12 row cultivator
7. 6 row no-till cultivator
8. Modified sprayer use for compost tea

Winter wheat and winter rye had already been planted in the fall of 2013. The spring arrived late in 2014 and there was a short planting window for early small grains. Oats, Barley, and Triticale were all planted on land that had been planted to corn in 2013. Corn and soybeans were planted next into tilled ground. Corn was planted into plowed down clover and the soybeans were planted on land that had been cover cropped with rye and then disc harrowed. The planting season ended with the planting of no-till organic corn and soybeans into rolled and crimped cover crops.

An organic fertilizer blend made of poultry litter, sulfate of potash, and Chilean nitrate was applied to all corn crops and the small grains. At planting, compost tea was injected adjacent to the corn and soybeans and mycorrhizal fungi were sprayed directly on the seed as an inoculant.

The spring small grain yields in 2014 were disappointing, amounting to 20 and 30 bushels per acre, and winter wheat yielded 34 bushels per acre. Corn and Soybeans yielded well. The organic no-till corn and soybeans yielded the best with no-till organic corn on rye and vetch cover crop yielding 161 bushels per acre and no-till soybeans on a rye cover crop yielding 42 bushels per acre. No-till corn yields on the rye and Austrian winter pea cover crop blend yielded considerably less at 102 bushels per acre.

All of the corn, soy, oats, triticale, and barley produced were saved for feed production and the winter wheat was sold onto the commodity market. After small grain harvest, large parts of the farm were planted to rye and vetch to prepare for no-till corn production and rye to prepare for no till soy production. Winter Wheat was planted following soybeans and some lighter ground set aside for barley.

In 2014 Stone House Farm purchased larger steers than usual with the objective of getting the steers to finished weight (ready for slaughter) on grass. Weight gains averaged two pounds per day and the cattle were sold in the winter of 2015 as finished steers.

2015

In 2015 all corn and soybeans were planted using organic no-till methods with the roller crimper on cover crops that were planted in 2015. The spring of 2015 came late and followed a very snowy and cold winter followed by lots of rains. This made for difficult soil conditions in the heavy clay and laid bare the challenges associated with farming in heavy clay soils. Spring grains were planted late and the corn had trouble getting off to a good start. An organic nitrogen side dress was used to help the corn crop recover.

In 2015 Corn yields were poor at 71 bushels per acre, but soy yields were good, at 40 bushels per acre (all in organic no - till conditions). Wheat yields were 35 bushels per acre. The best yields in 2015 came on well drained soils.

With the slow start to the 2015 growing season and poor small grain yields, Rick and I decided to prepare more land for 2016 tillage corn and we planted clover cover crops following some

small grains and rye vetch cover crops on lighter soils. Due to poor no-till yields in 2015 it was decided to continue no-till practices, but only on lighter soils.

The farm acquired a Case 290 tractor, and a Landoll disc ripper in 2015 to help with additional tillage. The Landoll was often a problematic tool, especially in heavy soils, where it caused compaction.

Grass fed finishing remained the focus of the grazing operation and 2015 was a pretty good year in terms of animal weight gain and sales. Rick did a good job managing the herd for weight gains and health.

Overall, the 2015 season was a challenging and difficult year with poor weather, low crop yields, and a difficult team that was unhappy farming organically.

2016

In January 2016 I took over management of the farm from Rick Burd and hired Kaspar Meier to help manage the fields.

Kaspar acquired a Plow Packer Pony (a plow connected to a soil conditioner and seeder) to do plowing, soil prep and seeding in one pass. Kaspar also sourced a swather and combine pick-up heads that the farm uses to this day to harvest weedy crops or crops that need to dry in a windrow before harvesting.

Additional lands were added to the operation in 2016 including the Esselstyn, and Runyon lands in Claverack, as well as, the Thorne and Davidson properties adjacent to Stone House Farm. The land was acquired with the goal of expanding grain production and the cattle grazing operation to achieve better economies of scale and meet the demand for grain.

2016 was the best organic crop year in all of the organic years at Stone House Farm so far. The plow packer pony was very effective at generating good crop yields on the lighter soils and roller crimping was used to grow successful soy and corn crops. Kaspar was a terrific operator and our collaboration resulted in very good organic crops. Unfortunately, Kaspar's personality was a difficult fit for the team and the operation and he was asked to leave before the season ended.

Corn yields averaged over 120 bushels across 400 acres, wheat yields were 55 bushels per acre, and soybeans averaged 48 bushels per acre with some fields yielding over 60 bushels per acre. These successful crops left the operation with a lot of grain to start the organic feed business and sell wholesale. Straw and first cut hay yields were also terrific in 2016 and filled the barns. These high yields enabled the growth of the farms hay business and straw sales

Skip McCagg joined the operation in 2016 and took Kaspar's role of field manager late in the season. This marked a change in practices and philosophy with more tillage, less organic no-till practices, and more equipment, often resulting in increased soil compaction and lower yields. Clarence Haywood joined the operation in late 2016 and became an important part of the team for the next four years.

The cattle did not gain weight as well in 2016, but prices for grass fat steers held steady.

Despite the higher crops yields and lower costs in 2016, the operation was unable to make a profit.

2017

In 2017 the farm continued to manage the extra acreage in Claverack and the Thorne and Davidson properties adjacent to Stone House Farm lands.

The following equipment was purchased:

- Case 250 tractor
- Valmar cover crop seeder - to mount on twelve row cultivator
- Robotic eye hitch (to control cultivator and minimize crop loss)
- Rollover plow
- Mixer grinders for feed production

In 2017 yields across the operation dropped by at least twenty percent. More disking was used and the plow packer pony was not used for any land preparation. There was not very good buy-in with the crew on organic no-till practices, and as a result, the no-till plantings were poorly executed.

Cover crop biomass in the spring of 2017 was poor and compaction increased across the farm with additional use of the Landoll and disc.

Hay yields continued to be strong in 2017 and I made the decision to begin expanding hay production in the rotation to help control weeds and build soil fertility.

Cattle prices dropped in 2017 and weight gains were disappointing. Late in 2017 I hired Kyle Dodge part time to assist with our cattle program to help improve the grazing rotation and weight gains.

2018

In 2018 the farm acquired more equipment in an attempt to maximize efficiency and improve crop yields.

The following equipment was acquired:

1. Chandler Poultry litter spreader
2. Rollover plow (to replace rollover purchased in the previous year)
3. Field cultivator (to improve soil conditions for planting)

The farm also acquired equipment from the Hudson Pines operation in 2018 including a skid steer, Case 110 tractor, and additional hay equipment.

It was decided to no longer farm the cropland in Claverack, and I handed the leases off to Larry Lunt's Armonia group. This land had been part of a strategy to achieve greater economies of scale and transition more land in the area to organic production, but it proved difficult managing these fields that were six to eight miles from the home farming operation with the reluctant crew I had in place.

Crop yields improved in 2018, especially corn and small grains. Soybeans continued to struggle with poor implementation of organic no-till practices and weed pressure. In 2018, large parts of the farm were under-seeded to hay with small grains, with different mixes used for heavy soils vs. lighter soils. This was in response to increasing weed pressure and soils

having been farmed in annual crops for over fifty years. The perennial hay crops (with appropriate mineral and fertility applications) have helped build carbon and nitrogen stocks in soils across the farm.

High-CBD hemp was grown for the first time on Stone House Farm on the home farm and the yields were good. The hemp revenue was particularly important to the farm in 2018 as there was no longer financial support from the New York office.

Cattle prices continued to decrease in 2018 and weight gains on the cattle from Virginia continued to be inconsistent. By the end of the season, I had decided to discontinue the grass-fed beef program and lease the pastures and facilities.

2019

The 2019 marked some significant changes in the operation. Hay production greatly increased, hemp production scaled up, pulse crops were planted following corn (peas, sunflowers, buckwheat), and renovation of the Hermitage dairy facility began, with the intent of integrating an A2-A2 grass-fed dairy operation on the farm.

The farm leased two additional Case tractors, a 280 and a 150 to help with additional hay production and replace a 25 year old Case 8940 tractor. A compost spreader was also purchased to spread manures and composts.

In 2019 the cattle pastures were leased to the Thorne's, and the Stone House beef cattle operations were discontinued.

In 2019, most crops yielded well (with the exception of soybeans and wheat) especially the new pulse & oilseed crops following corn that included peas and sunflowers. Hemp production was scaled to 32 acres and once again provided critical income to help keep the operation going.

2020

The 2020 season was the most difficult season since the transition. A drought lasted from May through August and resulted in poor corn and soy yields and failed pulse crops.

In 2020 even more of the farm was seeded down to hay and perennials as part of a plan to improve tired soils across the farm, limit grain production on marginal soils, and prepare the farm for a range of future possibilities. These seed-downs left Stone House Farm with large tracts of perennial forage & pasture, improving soil fertility, and controlling weed pressure.

Hudson Carbon

Background

In 2014, during the first full season of organic transition at Stone House Farm, Old Mud Creek Farm, Ben Dobson and Abby Rockefeller began discussing the possibility for implementing a monitoring/research project across the farm properties, with the goal of quantifying the change

in soil carbon and other soil health indicators over time as the transition progressed. The impetus for the idea was, at that time, an understanding and recognition of the fact that there was an opportunity within the context of the larger regenerative organic transition to build a robust, unique dataset that could demonstrate the efficacy of regenerative organic agricultural practices in building stocks of soil organic carbon.

Motivated by this opportunity, Abby & Ben began conversations with various academic institutions, non-profit research organizations, and independent experts, to aid in the development of the project, particularly the design & data collection aspects. This included Cornell, University of Massachusetts, The Rodale Institute, Tom Goreau, Martin Mitchner, and others.

Initial Research Design

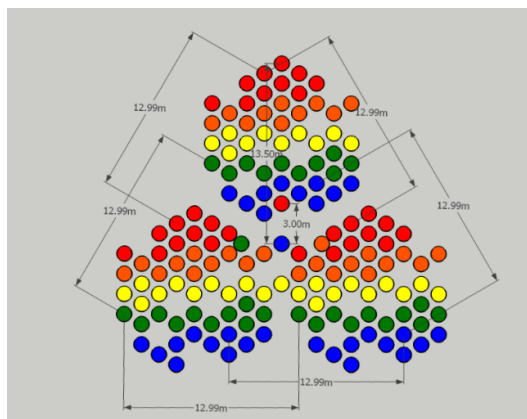
The initial research design was crafted around the core aspects of the farm management system, including historic management, coupled with the transition plan for how management would evolve across the farm. In addition, the diversity of soil types were layered on top of these management considerations in order to inform site selection. Historically, the farm had been partitioned into roughly two general management categories: perennial hay/pasture, and annual commodity cropping. The fields that fell in to the perennial category generally were in heavier clay soils, while fields that were continuously cropped were the lighter, sandy loam soils that were much more forgiving and productive. This was not a hard-and-fast rule, however, as there were outliers in both categories. There were also quite a few perennial pastures that are in rocky channery soils, and continue to be to this day. Looking forward from the perspective of crop rotation, part of the transition plan was to build out “full” rotations that included an annual phase, followed by a perennial phase, that could lead back into annual cropping, thereby adding a third general management category, i.e. annual/perennial.

Therefore, site selection for soil sampling was stratified across three different management categories that were being carried into the future via the organic transition plan, as well as three general soil types, including heavy silt/clay, light sandy loam, and rocky channery loam. Based on these strata, there were 12 original sites that were selected across Stone House, Old Mud Creek, and The Churchtown Dairy.

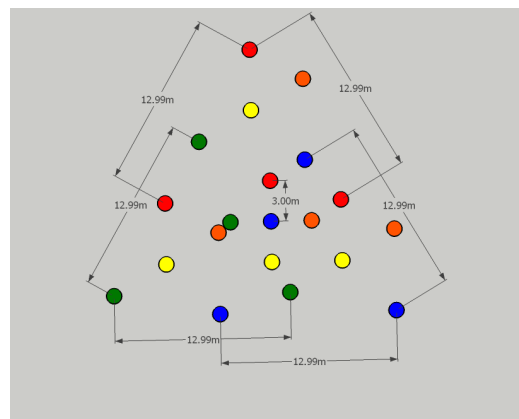
At the time the conversation around soil carbon in agricultural soils had not yet fully moved into the world of carbon credits and carbon offsetting, with considerations of additionality, permanence, total stocks not on Hudson Carbon’s radar. Therefore, in retrospect, the project design was limited in its capacity to accurately quantify baseline stocks at field and farm scale, as well as to quantify change over time in those stocks, in terms of tons of carbon per acre. In contrast, the question at hand leading the nascent Hudson Carbon project was more of a qualitative nature; the motivation that gave rise to the project was to understand, at a systems level, if a regenerative organic production that was based in diversity, diversification, tillage reduction, animal integration, and other agro-ecological hallmarks, could build and store more organic carbon throughout a deep soil profile over time than conventional chemical production.

In late 2014, into early 2015, the twelve monitoring sites were established across the three farm properties. As noted above, these sites were selected to be representative of the entire farm landscape, taking into consideration soil type and historic & future management.

C-Site	Soil Type
CS 1	Knickerbocker fine sandy loam
CS 2	Niagra silt loam
CS 3	Niagra silt loam
CS 4	Hoosic gravelly sandy loam
CS 5	Collamer silt loam
CS 6	Collamer silt loam
CS 7	Collamer silt loam
CS 8	Knickerbocker fine sandy loam
CS 9	Nassau channery loam
CS 10	Blasdell channery loam
CS 11	Blasdell channery loam
CS 12	Hoosic gravelly sandy loam



Triangular sample pattern showing 10 years of sample points



One Year's worth of sample points

The research design was ultimately geared towards monitoring change over time at the twelve research sites. Drawing on the work of Peter Donovan¹, we developed a triangular grid sampling pattern allowed for a set of 3 replicate points, one set per month for five months during the growing season, for 10 years, that are positioned around a fixed central point such that each set would be sampled only once over the ten-year period. Biomass was sampled once a month, and soil cores were extracted from the sample points after the first round of biomass collection, and following the last round of biomass collection, for a total of two samples per annum. The monthly biomass sample was a composite of each of the three sample points and

¹ Peter Donovan, "Measuring Soil Carbon Change: A Flexible, Practical, Local Method", 2013

is divided into two layers. The first layer we referred to as the *apogeal layer* which is the standing biomass. The second layer was referred to as the *epigeal layer* which is composed of prostrate biomass and surface litter. The apogeal layer was abbreviated as the A layer, and the epigeal as the B layer. The soil core sample was referred to as the C layer, which was then subdivided into horizons. This protocol allows us to measure the change in carbon stocks in biomass and soil over time. Our rationale for monthly biomass sampling was to test the hypothesis that biomass production was positively correlated with short-term changes in soil carbon.



Biomass transect



Biomass Transect, pre-harvest



Biomass transect, post A-layer harvest



Biomass transect, post B-layer harvest

In the first year of sampling (2015), our soil sampling methodology included monthly sampling to a depth of 45cm, using a slide-hammer with a 15cm sleeve to take three incremental core samples. In the fall of 2016, Hudson Carbon procured a tractor-mounted Giddings hydraulic soil coring machine, and began to pull full 1-m cores. There was a practical motivation for this equipment upgrade, which was that it was simply much easier to pull a full-length core with

hydraulic power, as opposed to multiple incremental cores using a slide-hammer. However, the research motivation behind extending the core length to 100cm was to discover how soil carbon dynamics change as depth increases, and how that differs between soil types. Because the crop rotation that was being developed at Stone House Farm at that point included a perennial phase on a significant portion of the fields, and one of our research sites (C-Sites) was in continuous hay production with perennial grasses, legumes & forbs, we hypothesized that there could potentially be non-trivial gains in soil carbon where perennial roots were venturing below 45cm. We continued with a biannual (spring-fall) coring schedule across the C-Sites through 2017, and switched to an annual schedule (fall/winter) in 2018.

Soil Analysis

Over the winter of 2015-2016, we began processing soil in-house per a set of protocols that were settled on after a series of conversations with the advisors & consultants noted above. This included 2mm, air & oven drying for bulk density, Loss-on-Ignition for per cent organic matter, dry sieving for soil texture, wet sieving and elutriation for texture analysis, and a Water-Stable Aggregation (WSA) test developed by Kris Nichols while she was at Rodale. In early 2016, the renovation of our on-farm laboratory was completed at Old Mud Creek Farm, and we continued our in-house analysis full-time. Soil core stratification evolved from three 15-cm sections to 5 sections, C1-C5, where:

C1 = 10cm (0-10 cm)
C2 = 10 cm (10-20 cm)
C3 = 10 cm (20-30 cm)
C4 = 30cm (30-60 cm)
C5 = 40 cm (60-100cm)

This depth stratification reflected what we hypothesized as high variability in carbon stocks in the top 30cm of soil, and decreasing variability moving down in the profile. After each individual replicate core was stratified per the above measurements, each layer (C1-C5) of each replicate was combined to create a composite. A portion of this soil was saved for in-house analysis, but we also sub-divided a portion of each composited layer to send away for outside lab analysis.

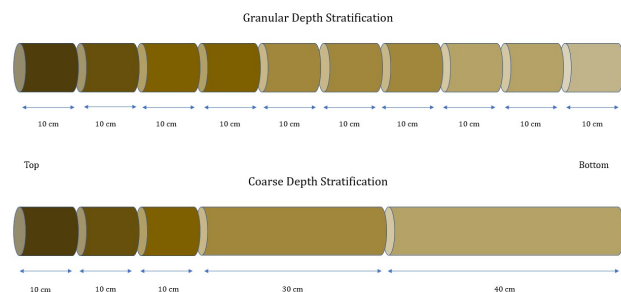
One challenge that we faced early on was that there were three C-Sites located at the Churchtown Dairy that were in the Blasdell and Nassau soil series, which are characterized by a channery loam soil type. These coarse channery fragments made gathering full intact cores extremely difficult. Therefore, we decided to forego gathering soil samples at these sites, and eventually ceased data collection entirely on two out of the three sites. Our third Eddy Flux tower is still located in the South Field at Churchtown, which was the location of CS-11.



Side-By-Side Systems Trial with Eddy Covariance

In 2018, we kicked off a multi-year study with Dr. Jim Tang at the Woods Hole Marine Biological Laboratory. The study was designed to compare the overall carbon and nitrogen balance between a regenerative organic cropping system and a conventional cropping system. Additionally, we chose a site at the Churchtown dairy that had been seeded down to a perennial forage mix a few years prior, to compare a site in continuous perennial with both annual cropping systems. For the annual cropping systems, we used the USDA-NCSS SSURGO and SATATSGO soil survey to choose sites that met the minimum area requirement for the Eddy Flux towers (150m radius), but that also shared a similar soil type, which is Knickerbocker fine sandy loam. The crop rotation carried out on each site was intended to be identical, with corn in year-1, soybeans in year-2, and spring barley in year-3. Management of each site was carried out at the discretion of the respective farm managers, which includes seed selection, tillage, fertility, crop protection (for conventional management), and cover cropping. The variation of the management approach was intended to represent the differences between the regenerative organic system of Stone House Farm and the management of the conventional site in terms of carbon (and nitrogen?) flux. In addition to eddy covariance data, we collected weekly chamber flux data at 6 random sites immediately around the tower, and collected post-harvest soil cores. The soil sampling design was a randomized selection of three points across the field, that were then composited by depth strata (C1-C5). We also conducted a nitrogen leaching study based on a methodology employed in another study carried out by Dr. Faming Wang, a member of Dr. Tang's research team. The data collection for this part of the study was inconsistent due to logistics and lack of training, and we therefore ended up with an incomplete dataset².

The crop rotation unfolded as intended, with corn in year-1, soybeans in year-2, and spring barley in year-3. The regenerative organic system had consistent cover in the form of cereal rye in between each cash crop. The spring barley was under-sown with a mix of alfalfa, orchard grass & brome, which established under the canopy of the barley crop, and filled out into a thick perennial stand post-harvest. There was heavy tillage (moldboard plow, field cultivator) ahead of the corn crop, but minimal tillage for the rest of the rotation, both in preparation for cash crops, as well as seeding covers. The conventional site practiced bare fallow in-between cash crops, and heavy tillage with a disc ripper ahead of planting. After year-3, the conventional site went back to soybeans, while the regenerative organic site stayed in alfalfa-grass mix for hay, ultimately representing the “full” (annual-



Depth stratification if 1-m soil core samples. The depth stratification we used initially was the “coarse” version, but we moved to the “granular” stratification in 2019.

² “Nitrogen Mineralization and Leaching in the Early Stages of a Subtropical Reforestation in Southern China”, Faming Wang, Weixing Zhu, Hanping Xia, Shenglei Fu, and Zhian Li, *Restoration Ecology Vol. 18 No. S2, pp. 313 – 322*.

perennial) rotation that was being established across much of stone house farm.

In 2019, the eddy covariance tower at the Churchtown Dairy was equipped with an open path methane sensor (LiCor Li-7700), which was intended to look at differences between field-scale methane flux during grazing periods and rest periods, and by correlating with chamber flux data, begin to understand overall methane balance within a perennial pasture. In fall of 2021, Jay Goldmark had officially taken over management of Stone House Farm, and we had at that point extended the original duration of the study by a full year. Jay asked that we remove all of our in situ equipment, so we had to conclude the study at that point, and we did likewise at the conventional site. However, we still have the third eddy covariance array (with methane sensor) deployed at the Churchtown Dairy.

Montana

In the summer of 2021, we had the opportunity to travel to the J Bar L Ranch in Montana's Centennial Valley to extend our soil carbon research into a rangeland context. The J Bar L ranch covers approximately 30,000 acres of deeded, state, & federal rangeland in southwest Montana, located in Madison County. With a foundation in Holistic Management, the ranch managers were interested in better understanding their rangeland soils within the context of carbon. One of the questions we set out to answer was simply around the feasibility of quantifying soil carbon empirically at that scale (10,000 ac.+).

At the time, project developers and ag tech companies were developing methodologies and tools to quantify soil carbon at scale. Regen network in particular had employed an approach to quantify soil carbon at their first project at Wilmot Station in NSW, Australia. Their approach integrated remote sensing, particularly multi-spectral imaging, as a way to inform sampling design as well as spatial interpolation for empirical modeling of farm/ranch scale carbon stock change. We had our doubts about the efficacy of their approach, and were interested in testing it out in Montana.

When endeavoring to determine a sampling design & methodology by which we could test the accuracy of Regen Network's protocol in this context, we were faced with the challenge of stratifying sample point selection across such a large area of interest, while keeping the sampling burden below an economic threshold. We therefore became interested in statistically-driven sample point stratification approaches, rather than stratified-random, or gridded-random approaches. It was our understanding that this would give us a relatively high degree of confidence that our sample set is capturing the spread of soil carbon across the sample area, while allowing us to keep the number of samples within an economically feasible range. We then concluded that cLHS was an approach that fit our needs in this context, while also being an approach that fit the capabilities of our team at the time.

The intent of our work in Montana ultimately pivoted away from testing the efficacy of the Regen Network protocol, and ended up being more about setting a baseline upon which to establish trends in soil carbon flux over time, should the interest and funding be available to do so. These decisions were primarily driven by the desires of the ranch owners and managers as the project evolved.

Reflections on Original Design, Most Recent Work

As noted above, a significant limitation of our original design was that it did not lend itself to interpolating soil carbon baseline stocks and stock change, and could not allow us to attribute changes in soil carbon stocks to individual practices, or systems of practices. However, it did yield data indicating trends in soil carbon dynamics over time at a whole system level, which could be valuable. Our decision to work with cLHS for statistically-driven soil sample stratification in the Montana rangelands context, along with our reflections upon the limitations of our original design, eventually led to an interest in applying it within the context of our Hudson Valley work.

We then applied cLHS for multiple different projects, including a full-farm stratification across the ~2000 acres of Stone House Farm, multiple high-density runs across Old Mud Creek, and for multiple collaborative projects, including two farms in Vermont, as well as with the Hotchkiss School, a private preparatory high school in Connecticut that runs a farming program on 130 acres. While we are still working through the soils for Vermont, and Old Mud Creek, we have processed and analyzed data for the Stone House and Hotchkiss projects, as well as the two different baselines we conducted in Montana. These data offer some interesting perspectives on the implications of scale and diversity of management practices, both spatially and temporally, which we are interested in unpacking further.

From our perspective, our entire set of cLHS data could answer some interesting, poignant questions related to the efficacy of cLHS as a stratification approach for empirically-based soil carbon stock estimation, and could contribute to the continually emerging literature focused on assessing different sample stratification approaches, including K-Means, stratified random, and gridded sampling.